AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 24 · TEACHER GUIDE

Train, Validation, Test

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach roles by information use and source identity.

Supply all promised actions,declared tie policies and complete group ranges without inventing final outcomes.

Keep one habit: Require development exposure,source counts and freeze status beside the selected candidate.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary target: Select a model using a declared **validation** rule

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Teach two 45-minute sessions. This week focuses on **validation** and model selection. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Validation can guide model,feature,setting or threshold choices. Those decisions adapt to its evidence,so preserve a separate final set for the frozen selected system and report exposure honestly.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Saved classroom project records for critique; no personal student dataset required

Opening retrieval

Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

Look for: Freeze the full evaluation procedure before testing

Reference example, not the only acceptable wording: Teacher explanation: Accuracy 15/20, precision 3/5, recall 3/6. Report TN 12 FP 2 FN 3 TP 3 and blocked positive.

Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate MAE MSE and RMSE

Reference example, not the only acceptable wording: Teacher explanation: (1+2+3)/3=2 cm.

Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

Look for: Categorize development errors using **evidence**

Reference example, not the only acceptable wording: Teacher explanation: Dim 4/12 about 33.3 percent; bright 3/18 about 16.7 percent.

End goal

By the end, I can explain train, **validation**, test using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Hold Out Test Data

Absolute Error and MAE

Error Analysis

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

You have 30 leaves with 3 views each. Allocate 18 leaves to train,6 to **validation**,6 to test. How many images are in each? Where is a learned normalisation mean fitted?

Concept to explain

Training fits parameters, **validation** chooses settings, and testing estimates final performance after choices are frozen.

Term	Meaning
training set	data used to fit parameters
validation set	data used to choose settings
test set	data used for final estimate
split strategy	rule assigning examples to sets

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach fitting,selection and reporting as three separate jobs

> Use the stated validation metric and tie rule without consulting final outcomes.

Role definitions and complete main results

Fitting and selecting answer different questions. **Training** adjusts parameters. **Validation** helps choose settings or model families. A final test estimates the performance of the frozen development **process** on held-out examples.

Keep related records in the same split when separate rows could reveal the same source.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **validation** and model selection.

Candidate	Supplied training MAE cm	Supplied validation MAE cm	Declared selection fact
A	1	3	Simpler than C under the declared tie policy
B	0	5	Perfect training score does not decide selection
C	2	3	Ties A on validation, loses declared simplicity tie

Three candidates have **training** MAE A=1, B=0, C=2 and **validation** MAE A=3, B=5, C=3. The declared rule is smallest **validation** MAE, then simpler model, with A simpler than C. Choose A.

B fits **training** perfectly but does worse on **validation**. The final test remains unopened.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Term	Meaning
Validation	Development evidence used to choose between options
Hyperparameter	A model setting chosen outside its fitted parameters
Selection rule	A declared method for choosing an option
Test set	Held-out evidence reserved for a final evaluation

Validation and model selection at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Training	Fit each candidate	No test labels
Validation	Choose A by rule	Record every attempt
Final test	Frozen model only	Report after selection

A tie policy prevents a convenient choice after inspecting results. Equal validation scores do not prove equal future performance. Record candidate settings, split identifiers, preprocessing and the full comparison, including rejected options.

Where should parameter fitting, setting selection and final reporting occur? Explain the steps or evidence behind your response.

Read separate pictured result table

Assign a separate job to **training**, **validation** and final test **evidence**.

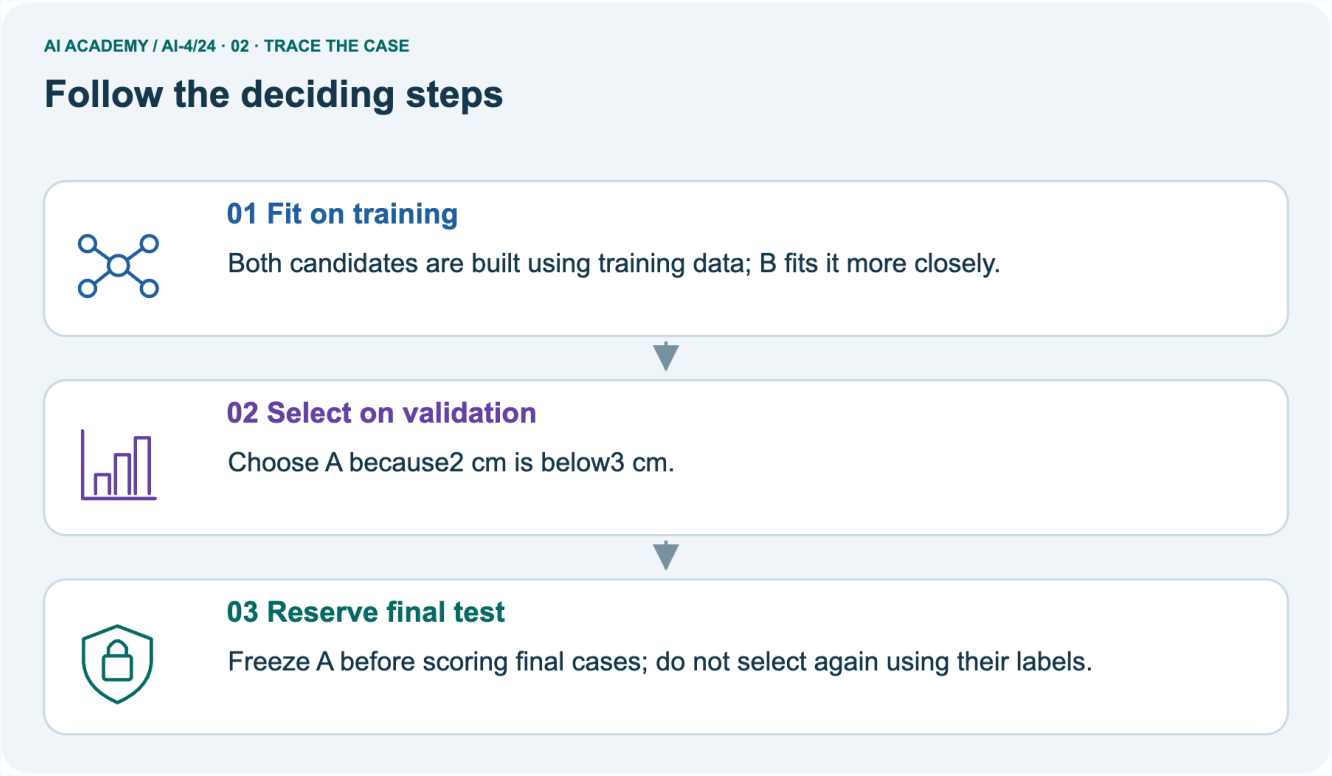
Candidates A and B have **training** MAE1 and 0 cm; **validation** MAE2 and 3 cm. **Selection rule** is smallest **validation** MAE.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Main A/B/Cresults choose A at validation MAE 3 cm, tied with C but simpler; Bhas training 0 and validation 5. Separate picture compares only A/Bwith validation 2/3 cm, again choosing A. The pictured change ties both at2 cm; predeclared simplicity still chooses A. These are supplied development results, not training code or final-test scores.

A lower inspected validation number does not guarantee population performance, and repeated choices can adapt to the validation sample.

Trace selection,freeze and changed tie



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Fit on training	Both candidates are built using training data; B fits it more closely.
2. Select on validation	Choose A because2 cm is below3 cm.
3. Reserve final test	Freeze A before scoring final cases; do not select again using their labels.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Freeze A before scoring final cases; do not select again using their labels. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Repeated **validation** use can itself overfit model choices; keep a genuinely untouched final evaluation.

Change: A and B both have **validation** MAE2 cm; A is simpler under a predeclared tie rule.

Prediction: Select A using the predeclared simpler-model tie-break.

Reason: The **validation** errors tie, so the stated secondary rule decides without consulting final data.

▼ **Additional100-leaf role example**

Split 100 independent leaves into 60 train,20 **validation**,20 test.

Explain roles and pre**process**ing.

Train fits parameters and learned transforms; **validation** chooses settings; final test evaluates the selected system. Group related records first.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Assign every modeling decision and dataset use to the correct split.

MODEL IDEA **Training** fits parameters, **validation** chooses settings, and testing estimates final performance after choices are frozen.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Fit two candidate trees on **training** rows. Choose depth and threshold using **validation** metrics and error slices.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 Fit two candidate trees on **training** rows.

STEP 2 Choose depth and threshold using **validation** metrics and error slices.

STEP 3 Freeze pre**process**ing, features, model, and decision threshold.

STEP 4 Evaluate once on test; do not return to tune and still call it final.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

Exact explanation to model

Explain the added worked example

Key terms

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

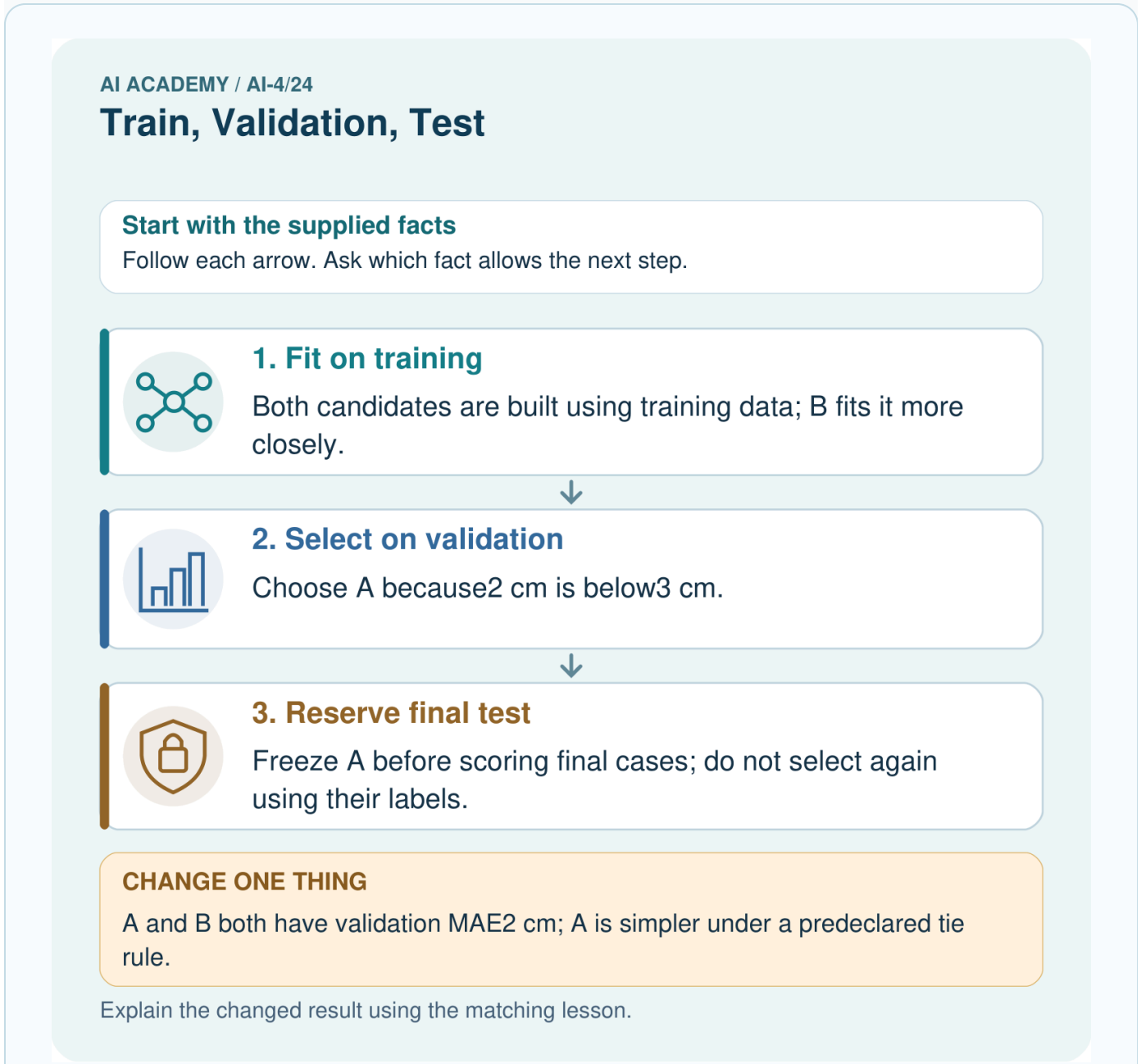
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Train, Validation, Test

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Guide the calculation, then remove help

> Collect each learner’s prediction before showing its key.

Current completion task and checked key

Completion task: Where should parameter **fitting**, setting selection and final reporting occur?

Teacher check: Teacher explanation: Fit on **training**, select on **validation**, evaluate the frozen choice on held-out test; fit pre**process**ing only on **training**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Validation** and model selection. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: Choose A: **validation** MAE 3 ties C, and A wins the simplicity tie. B has **validation** MAE 5 despite **training** MAE 0.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

Why does B's zero **training** MAE not settle the original choice? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Model selection is based on held-out **validation** error here. Lower **training** error can reflect **fitting** the **training** details rather than better new-case **prediction**.

Changed-case check: Select A using the predeclared simpler-model tie-break. The **validation** errors tie, so the stated secondary rule decides without consulting final data.

Changed-case reasoning: Select A using the predeclared simpler-model tie-break. The **validation** errors tie, so the stated secondary rule decides without consulting final data.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: Split 100 independent leaves into 60 train,20 **validation**,20 test.

Write the question to explain: Explain roles and pre**process**ing.

Reveal after **prediction**s: Train fits parameters and learned transforms; **validation** chooses settings; final test evaluates the selected system. Group related records first.

Use the earlier transfer question as additional practice. Reserve the selected fresh P/Q/R**validation**-and-test-action task for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided response: Fit on **training**, select on **validation**, evaluate the frozen choice on held-out test; fit pre**process**ing only on **training**.

Exit response: Selection used this sample, repeated attempts can adapt to it, and future conditions and sampling may differ.

Case-specific teaching sequence

Stage · Teacher action

> **Stage:** Session 1 start to 7 minutes
>
> **Teacher action:** Use the prerequisite question: Which data fitted the line last week? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify validation in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: choose a candidate; assign data roles. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Connect each conclusion to the stated evidence.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between validation and hyperparameter. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: A team creates D after reading validation mistakes. What changes in the record?

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Select a model using a declared validation rule; Keep training, validation and test roles separate. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Keep predictions, observed or supplied outcomes and unmeasured cases separate.

Supplied inputs and checked guidance

Role actions 1–6 trainingfit, 7–13 validation selection/development, 14 final preparation, 15–17 final evaluation/reporting, 18–20 invalid claims/boundaries needing repair. Manifest18/6/6 leaves and 54/18/18 images, no overlapping source groups. The allocation is illustrative. Selected Ptrains2, Q0, R1 butvalidationP4, Q6, R4; choose Punder declared simpler-than-Rtie.

Freeze all settings/reporting rules, then use reserved final evidence; no test score is supplied to choose again.

Predict, investigate and record

1. 1. Write the hypothesis and expected **evidence** before running.
2. 2. Role Assignment: sort twenty actions into train/**validation**/test, then group related examples before splitting.
3. 3. Run one normal, one boundary, one missing, and one shifted case.
4. 4. Calculate the metric from raw counts or row-level errors.

5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the experiment ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the current misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: Choose A: **validation** MAE 3 ties C, and A wins the simplicity tie. B has **validation** MAE 5 despite **training** MAE 0.

A stronger claim makes a stronger project: Selection used this sample, repeated attempts can adapt to it, and future conditions and sampling may differ.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: Choose P with **validation** MAE 4; Q loses with 6.

Keep the test unopened until settings and reporting rules are frozen.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve the selected fresh P/Q/Rvalidation-and-test-action task for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

MISCONCEPTION TO REPAIR Three random folders are enough even when photos of one plant appear in each.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected experiment transfer

> Keep this key until after a first attempt; record any help and prior exposure.

Selected independent task and checked key

Selected case: **Training** errors P 2 Q 0 R 1; **validation** errors P 4 Q 6 R 4. Rule: smallest **validation** MAE, then simpler; P is simpler than R. Choose and state the test action.

Teacher explanation: Choose P with **validation** MAE 4; Q loses with 6. Keep the test unopened until settings and reporting rules are frozen.

Judge trace, scope and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Choose a candidate

Apply the worked **selection rule**. Explain why B is not chosen.

Teacher explanation: Choose A: **validation** MAE 3 ties C, and A wins the simplicity tie. B has **validation** MAE 5 despite **training** MAE 0.

W2 Assign data roles

Where should parameter **fitting**, setting selection and final reporting occur?

Teacher explanation: Fit on **training**, select on **validation**, evaluate the frozen choice on held-out test; fit pre**process**ing only on **training**.

W3 Record another attempt

A team creates D after reading **validation** mistakes. What changes in the record?

Teacher explanation: Add D and its motivation to development history. **Validation** influenced D and is not untouched test **evidence**.

W5 Limit a claim

Why is the winning **validation** score not a guaranteed future score?

Teacher explanation: Selection used this sample, repeated attempts can adapt to it, and future conditions and sampling may differ.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Train fits parameters and learned transforms; **validation** chooses settings; final test evaluates the selected system. Group related records first.

Additional transfer practice question

Assign roles: set A fits model parameters; B selects between two models; C is opened once after choices are fixed. Which is train, **validation** and test?

Reasoned practice answer

A is **training**, B is **validation** and C is final test. Keep C separate from **fitting** and model selection.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: "**Validation** and final test data have interchangeable purposes during development." Keep the first attempt and an uncoached retry separate.

Original transfer answer

Image counts are 54,18,18. Fit the mean on **training** images only, then apply the fixed transform to **validation** and test. **Validation** chooses settings; final test evaluates the selected system.

Do not compute the mean from all 90 images.

Explanations for the additional workbook examples

W1 Choose a candidate

W2 Assign data roles

W3 Record another attempt

W4 Apply independently

Training errors P 2 Q 0 R 1; **validation** errors P 4 Q 6 R 4. Rule: smallest **validation** MAE, then simpler; P is simpler than R. Choose and state the test action.

Teacher explanation: Choose P with **validation** MAE 4; Q loses with 6. Keep the test unopened until settings and reporting rules are frozen.

W5 Limit a claim

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Publish the grouped **split strategy** and counts for the Plant Classifier, including why plant identity stays within one split.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **validation** and **hyperparameter** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: Choose P with **validation** MAE 4; Q loses with 6. Keep the test unopened until settings and reporting rules are frozen. Fresh retry: **validation** L 2 M 2 N 3, simplicity M before L.

Choose M under the declared tie policy; do not use test scores to break the tie. Mark each criterion as independent, supported or not yet demonstrated.

Reassess the reasoning after feedback; do not replace the first attempt silently.

Delayed return

Suggested return points in this level: ai-4/25, ai-4/27, ai-4/30. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 25: Metrics Match Consequences. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week 25: Metrics Match Consequences. Carry forward exact split roles,group assignments,selection/tie rules and the recorded development history. Final testing was introduced in Week 23;repeated development and overfitting are revisited later.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching target: Select a model using a declared **validation** rule

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Sections 1–6; record the brief recall in the workbook.

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> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

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> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Where should parameter **fitting**, setting selection and final reporting occur?

Reasoned practice answer: Teacher explanation: Fit on **training**, select on **validation**, evaluate the frozen choice on held-out test; fit pre**process**ing only on **training**.

Check every learner before discussion: Use the worked case for **Validation** and model selection. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: Choose A: **validation** MAE 3 ties C, and A wins the simplicity tie. B has **validation** MAE 5 despite **training** MAE 0.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: A and B both have **validation** MAE2 cm; A is simpler under a predeclared tie rule. Require a **prediction** and a reason before comparing. Expected result: Select A using the predeclared simpler-model tie-break.

Reason: The **validation** errors tie, so the stated secondary rule decides without consulting final data.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

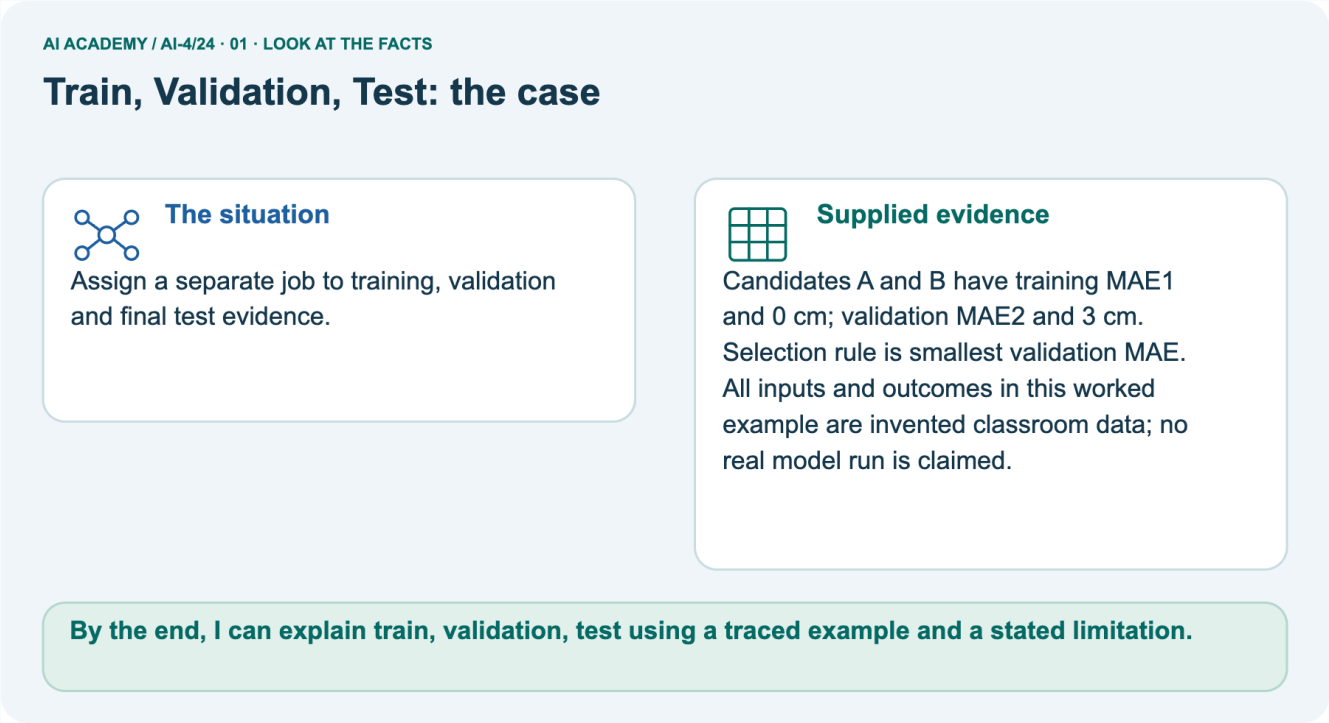
<details>
<summary>Reference · Teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Train, Validation, Test: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

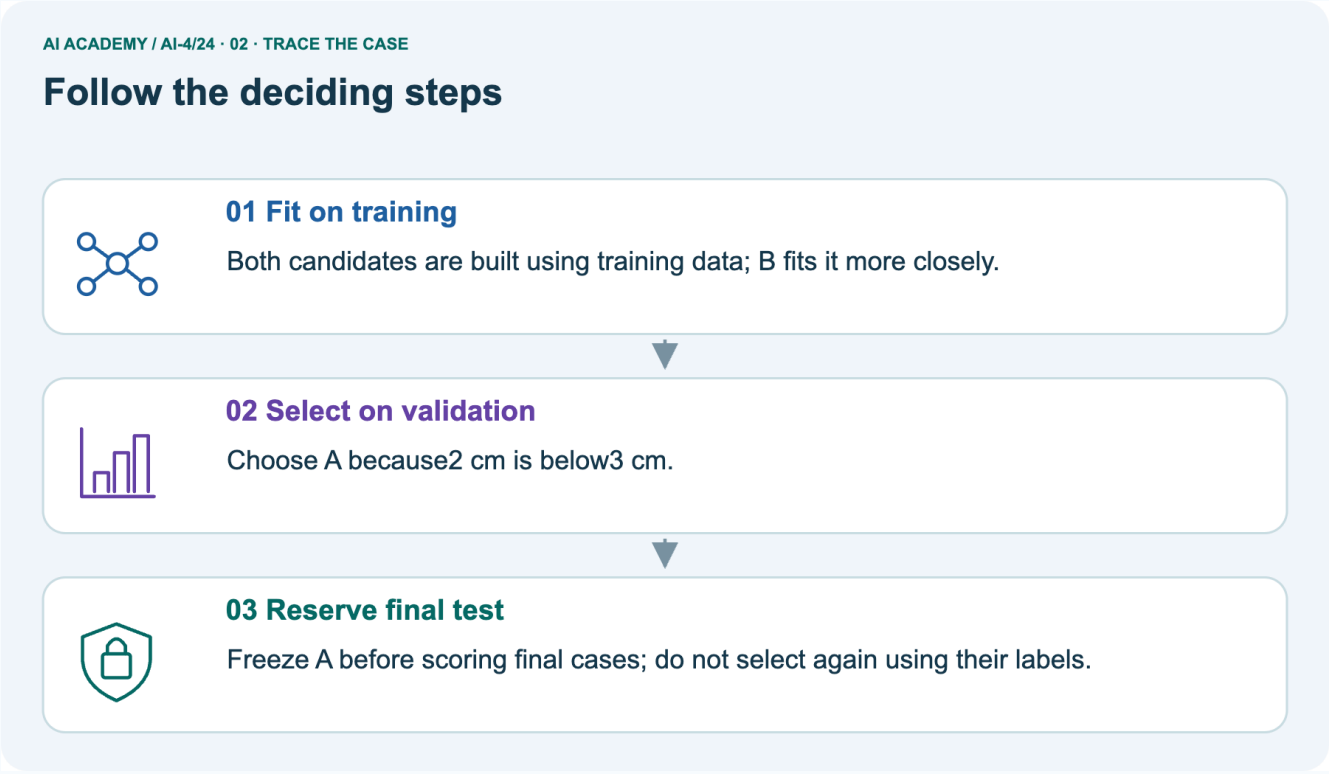
Assign a separate job to **training**, **validation** and final test **evidence**. Supplied facts: Candidates A and B have **training** MAE1 and 0 cm; **validation** MAE2 and 3 cm. **Selection rule** is smallest **validation** MAE.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



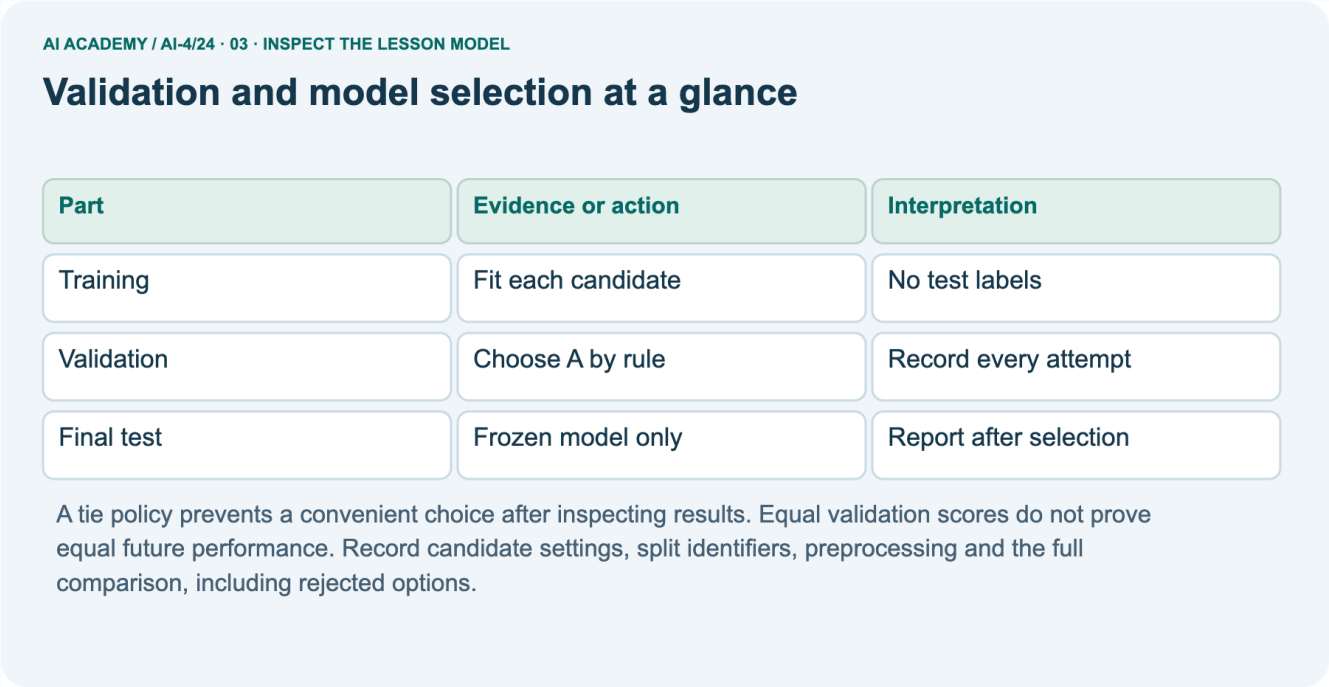
Follow the labelled picture with the matching explanation. Open the vector diagram

Fit on **training**: Both candidates are built using **training** data; B fits it more closely. → Select on **validation**: Choose A because2 cm is below3 cm. → Reserve final test: Freeze A before scoring final cases; do not select again using their labels.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Validation and model selection at a glance



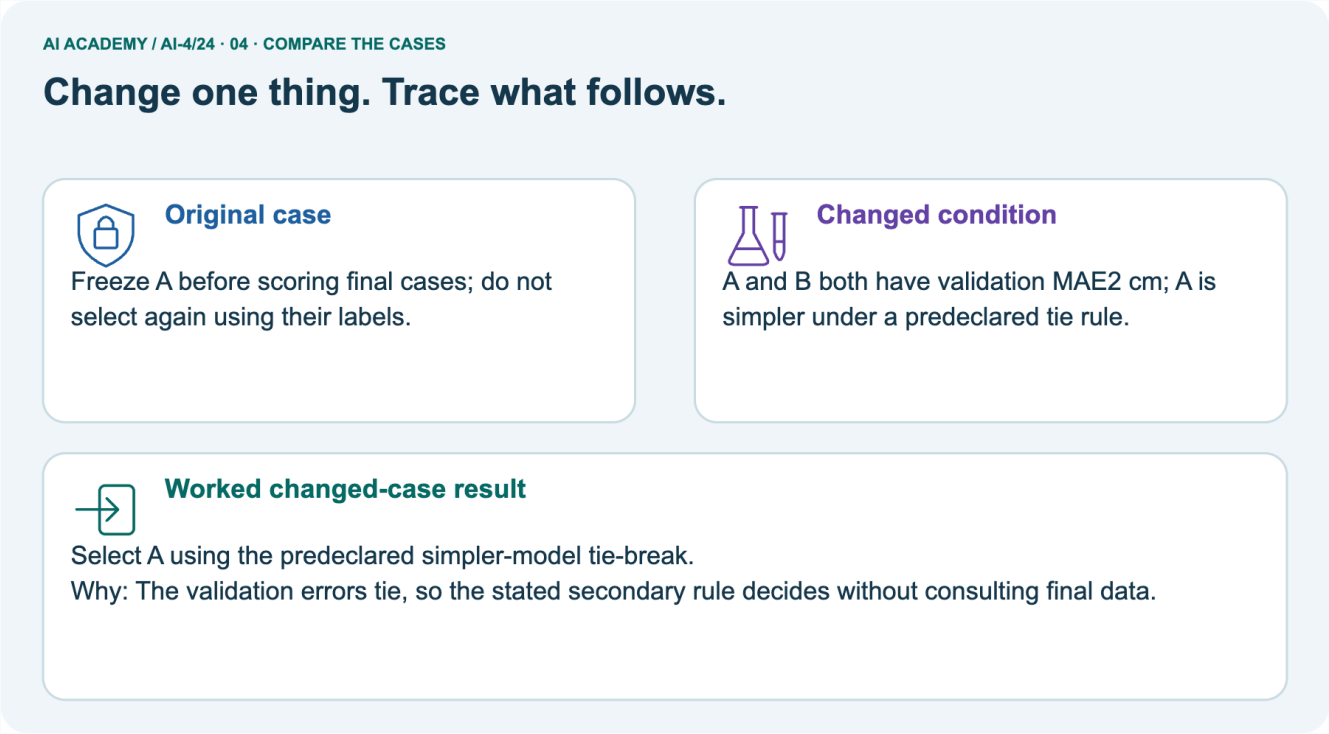
Follow the labelled picture with the matching explanation. Open the vector diagram

Validation and model selection at a glance. **Training** | Fit each candidate | No test labels. **Validation** | Choose A by rule | Record every attempt. Final test | Frozen model only | Report after selection

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Freeze A before scoring final cases; do not select again using their labels. Changed condition: A and B both have **validation** MAE2 cm; A is simpler under a predeclared tie rule.

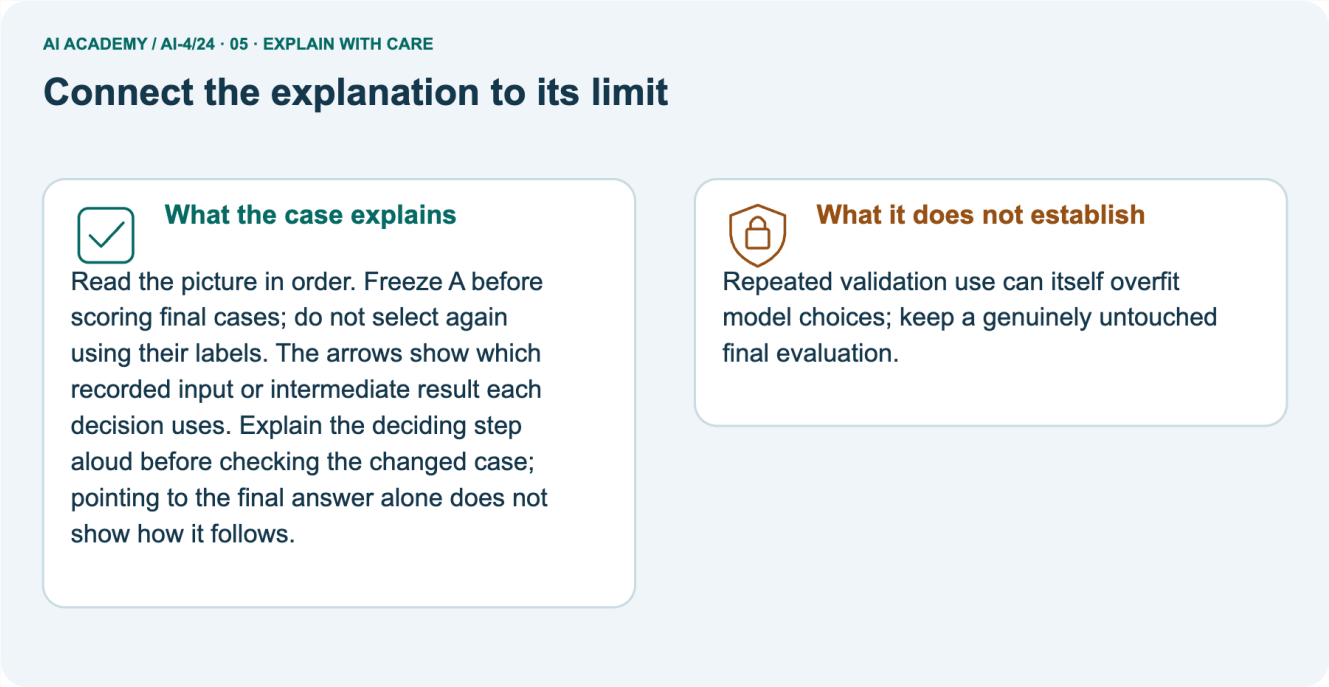
Expected result: Select A using the predeclared simpler-model tie-break. Reason: The **validation** errors tie, so the stated secondary rule decides without consulting final data.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The **validation** errors tie, so the stated secondary rule decides without consulting final data.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Freeze A before scoring final cases; do not select again using their labels. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

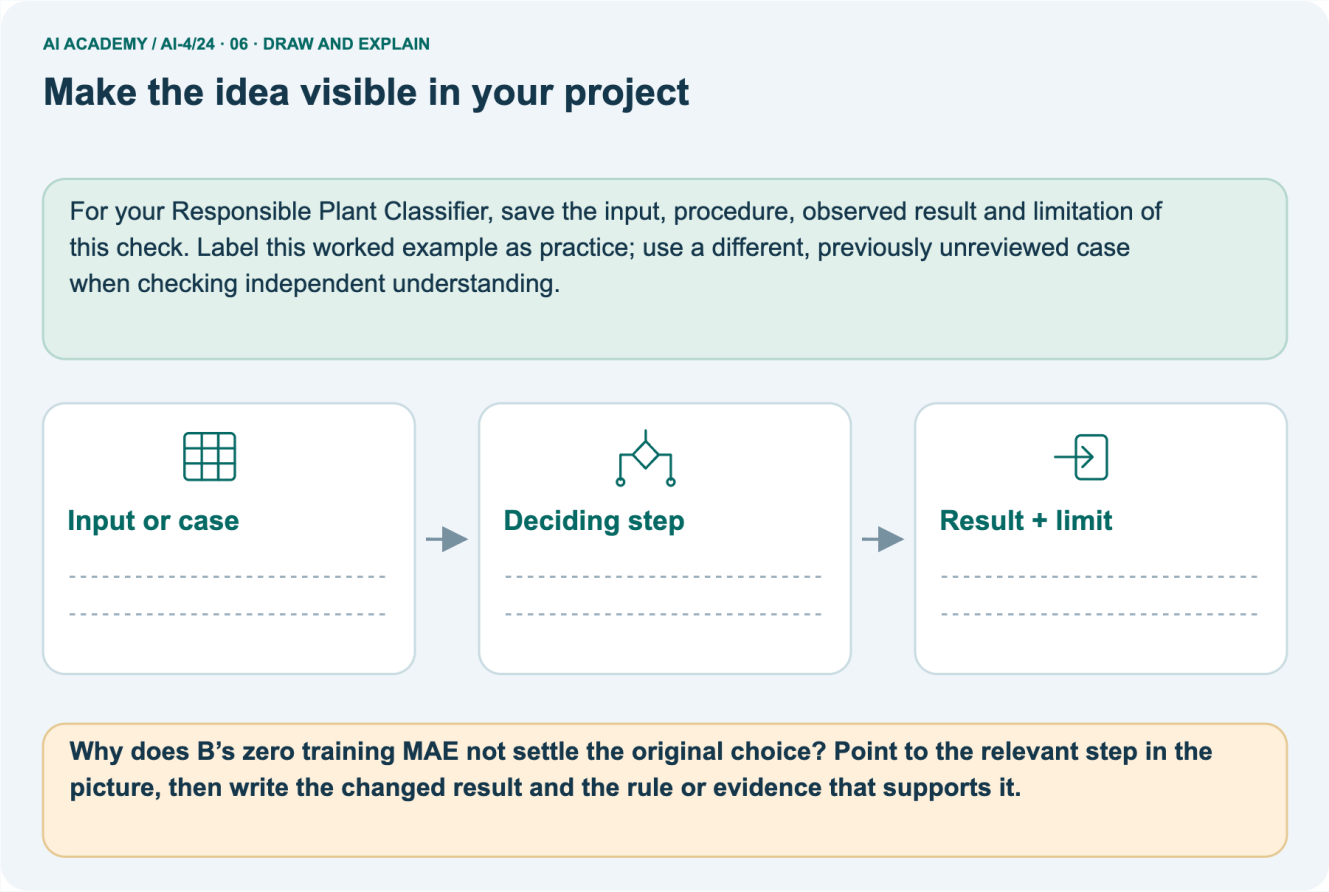
Limit: Repeated **validation** use can itself overfit model choices; keep a genuinely untouched final evaluation.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Repeated **validation** use can itself overfit model choices; keep a genuinely untouched final evaluation.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why does B's zero **training** MAE not settle the original choice? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Model selection is based on held-out **validation** error here. Lower **training** error can reflect **fitting** the **training** details rather than better new-case **prediction**.

Changed-case check: Select A using the predeclared simpler-model tie-break. The **validation** errors tie, so the stated secondary rule decides without consulting final data.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Train, Validation, Test: the case

02. Follow the deciding steps

03. Validation and model selection at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.